

ITM Tracing Configuration Summary

STM32F4 Project Setup

September 17, 2024

1 System Clock Configuration (CubeMX)

- **System Clock Mux:** Set to **PLLCLK**.
- **PLL Source Mux:** Set to **HSI** (High-Speed Internal clock).
- **PLL Configuration:**
 - **PLLM:** /8 (to divide HSI 16 MHz down to 2 MHz).
 - **PLLN:** x168 (to multiply 2 MHz up to 336 MHz).
 - **PLLP:** /4 (to divide 336 MHz down to 84 MHz for SYSCLK).
- **Final System Clock (SYSCLK):** 84 MHz.

2 ST-Link Debugger Configuration (uVision)

- **Core Clock:** Set to 84 MHz.
- **Trace Clock:** Set to 84 MHz.
- **SWO Clock:** Set to 12 MHz (to match the system).
- **ITM Port 0:** Enabled for **ITM tracing**.

3 Pin Configuration

- **PB3:** Configured for **SW0** by selecting **Trace Asynchronous SW** in CubeMX.
- **PA14:** Automatically set as part of the **SWD** interface.

4 ITM Tracing

- **ITM_SendChar():** Currently being used for output instead of **printf**.
- **MicroLib:** Enabled to ensure compatibility and proper operation of the system.